Title

Crop Recommendation System using Machine Learning

Aim

To design and analyze a machine learning–based crop recommendation system that predicts whether a farmer should grow **Rice, Wheat, or Maize** based on **soil nutrients and climatic conditions** using a crop recommendation dataset.

Objectives

1. To study the impact of soil nutrients **Nitrogen (N), Phosphorus (P), and Potassium (K)** on crop growth.
2. To analyze climatic parameters such as **temperature, humidity, and rainfall**.
3. To define the **input–output mapping** for crop prediction.
4. To identify the **machine learning problem type**.
5. To perform **exploratory data analysis (EDA)** using statistical and mathematical methods.
6. To prepare the dataset for building an accurate crop prediction model.

Theory

Crop Recommendation System

A crop recommendation system is an intelligent decision-support system that assists farmers in selecting the most suitable crop based on **soil fertility and climate conditions**. Machine learning techniques analyze historical agricultural data to identify patterns and relationships between soil parameters and crop yield.

Dataset Description

The crop recommendation dataset contains the following attributes:

Feature	Description
N	Nitrogen content in soil
P	Phosphorus content in soil
K	Potassium content in soil
Temperature	Average environmental temperature (°C)
Humidity	Relative humidity (%)
Rainfall	Annual rainfall (mm)
Label	Recommended crop (Rice, Wheat, Maize)

Input–Output Mapping

Input Vector (X):

$$X = [N, P, K, Temperature, Humidity, Rainfall]$$

Output (Y):

$$Y = Crop\ Type \in \{Rice,\ Wheat\ Maiz$$

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Machine Learning Problem Type

- **Learning Type:** Supervised Learning
- **Problem Type:** Multi-class Classification
- **Reason:**
    - Output is categorical
    - Multiple crop classes exist

Exploratory Data Analysis (EDA)

EDA is used to understand the dataset before model training.

Statistical Measures Used

1. **Mean**

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

2. **Standard Deviation**

$$\sigma = \sqrt{\frac{1}{n}\sum(x_i - \mu)^2}$$

3. **Minimum and Maximum**

$$Min = min(x), Max = max(x)$$

4. **Correlation Coefficient**

$$r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

These measures help analyze nutrient distribution, climate variability, and feature relationships with crops.

Statistical Measures (Step-by-Step)

Given Data (Sample Nitrogen Values in Soil)

Let Nitrogen content (N) for 5 samples be:

$$x = [40,\ 50,\ 60,\ 50,\ 70]$$

Number of observations:

$$n = 5$$

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1. Mean (μ)

Formula

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Calculation

$$\mu = \frac{40+50+60+50+70}{5} \mu = \frac{270}{5} = 54$$

Result

$$\mu = 54$$

Explanation

The mean represents the **average Nitrogen content** in the soil samples.

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2. Standard Deviation (σ)

Formula

$$\sigma = \sqrt{\frac{1}{n}\sum(x_i - \mu)^2}$$

Step 1: Calculate Deviation from Mean

x_i	$x_i - \mu$	$(\mu)^2$
40	40 − 54 = −14	196
50	50 − 54 = −4	16
60	60 − 54 = 6	36
50	50 − 54 = −4	16
70	70 − 54 = 16	256

Step 2: Sum of Squared Deviations

$$\sum(x_i - \mu)^2 = 196 + 16 + 36 + 16 + 256 = 520$$

Step 3: Divide by n and take square root

$$\sigma = \sqrt{\frac{520}{5}}\sigma = \sqrt{104} \approx 10.19$$

Result

$$\sigma \approx 10.19$$

Explanation

Standard deviation shows **how much Nitrogen values vary** from the average. Higher σ → more variation in soil fertility.

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3. Minimum and Maximum

Formula

$$Min = min(x), Max = max(x)$$

Calculation

$$Min = 40 Max = 70$$

Result

$$Min = 40,\ Max = 70$$

Explanation

- Minimum = lowest Nitrogen level
- Maximum = highest Nitrogen level
  These values show the **range of soil nutrient availability**.

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4. Correlation Coefficient (r)

Given Two Variables

Let:

- X= Nitrogen (N)
- Y= Rainfall (mm)

Sample	X (N)	Y (Rainfall)
1	40	120
2	50	140
3	60	160

Sample	X (N)	Y (Rainfall)
4	50	150
5	70	180

Step 1: Find Mean of X and Y

$$\mu_X = 54$$

$$\mu_Y = \frac{120+140+160+150+180}{5} = 150$$

Step 2: Calculate Covariance

X	Y	$X - \mu_X$	$Y - \mu_Y$	Product
40	120	−14	−30	420
50	140	−4	−10	40
60	160	6	10	60
50	150	−4	0	0
70	180	16	30	480

$$Cov(X,Y) = \frac{420+40+60+0+480}{5} = \frac{1000}{5} = 200$$

Step 3: Use Correlation Formula

$$r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Assume:

$$\sigma_X = 10.19,\ \sigma_Y = 22.36 r = \frac{200}{10.19 \times 22.36} r = \frac{200}{227.8} \approx 0.88$$

Result

$$r \approx 0.88$$

Explanation

- $r = 0.88 \rightarrow$ **Strong positive correlation**
- As **rainfall increases**, Nitrogen requirement also increases
- Helpful for **crop recommendation decisions**

Algorithm

Algorithm: Crop Recommendation using Machine Learning

1. Start
2. Load the crop recommendation dataset.
3. Identify input features:
   N, P, K, Temperature, Humidity, Rainfall
4. Identify output label:
   Crop Type
5. Perform data preprocessing:
   - Check missing values
   - Normalize numerical features if required
6. Perform Exploratory Data Analysis:
   - Calculate mean, standard deviation, min, max
   - Analyze crop distribution
   - Study correlation between features
7. Split dataset into training and testing sets.
8. Train a suitable classification model.
9. Predict the recommended crop for given input conditions.
10. Evaluate model performance.
11. Stop

BEGIN

1. Load Crop Recommendation Dataset

2. Define Input Features:

Nitrogen (N)

Phosphorus (P)

Potassium (K)

Temperature

Humidity

Rainfall

3. Define Output Label:

Crop Type {Rice, Wheat, Maize}

4. Read dataset into memory

5. Perform Data Preprocessing:

IF missing values exist THEN

Handle missing values

END IF

6. Perform Exploratory Data Analysis (EDA):

Calculate Mean of each feature

Calculate Standard Deviation

Find Minimum and Maximum values

Analyze crop class distribution

Compute correlation between features

7. Split dataset into:

Training set

Testing set

8. Train Machine Learning Classification Model

Using training dataset

9. Provide new soil and climate input values

10. Predict Crop Type using trained model

11. Display recommended crop to user

END

Conclusion-write in your own words: